

EXPERIMENTAL ANALYSIS AND COMPENSATION OF MOTOR – INDUCED MAGNETIC INTERFERENCE IN UAVs USING EMBEDDED SYSTEM



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Bachelors of Aeronautical engineering

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PROJECT GUIDE

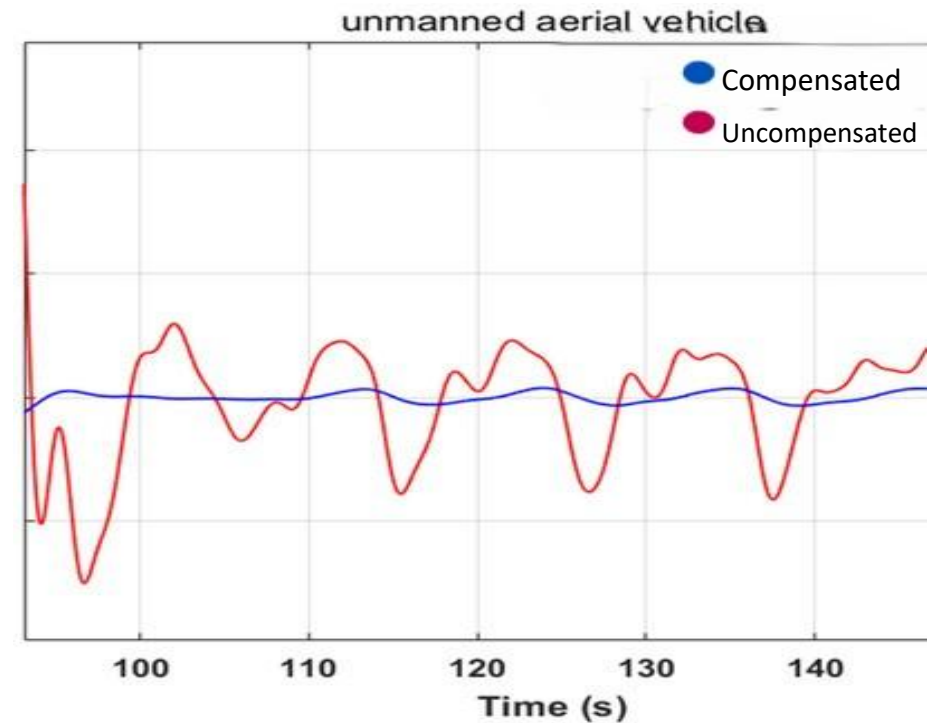
DR.V.GIRIDHARAN

OUTLINE

- Objective
- Introduction
- Literature Survey
- Problem statement
- Existing system
- Proposed system
- Methodology
- Components
- Circuit Design
- System Assembly
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- Summary

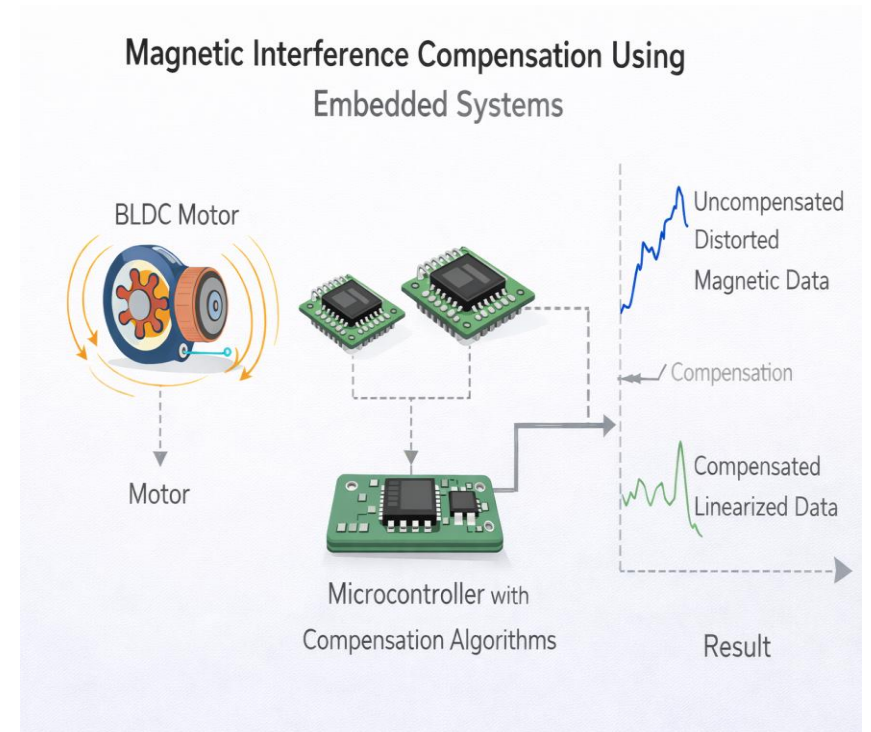
OBJECTIVE

- To **detect** and **measure** magnetic field **variations** using **sensors**.
- To **analyse** effects of motor-induced **interference**
- To **implement embedded algorithms** for real-time compensation.
- To **linearize magnetometer** data by compensating interference effects.



INTRODUCTION

- UAV navigation uses magnetometer
- Measures Earth's magnetic field for heading
- UAV **magnetometers** face **interference** from **drone components**.
- Magnetic **interference** could be 10 to 100 times **stronger** than the subtle **geological signals**.
- Embedded system collects real-time data.
- **Software compensates interference mathematically.**
- **Improves** UAV reliability for **geophysical surveys**.



LITERATURE SURVEY

□ **Bingyang Chen et al. — Magnetic Interference Analysis and Compensation Method of Airborne Electronic Equipment in a UAV (*Appl. Sci.* 2023) –**

- Analyzes magnetic interference from UAV
- proposes a linear compensation model based on power supply current (T-L model)
- Limitations - does not handle non-linear or rapidly changing interference

<https://www.mdpi.com/2076-3417/13/13/7455>

□ **Su Zhenning et al. — Aeromagnetic Compensation Method Based on Ridge Regression Algorithm (*Chinese Journal of Geophysics* 2022)**

- Applied ridge regression(keep coefficients small)
- addressing multicollinearity
- Achieved more stable result than T-L model

<https://www.sciopen.com/article/10.3969/j.issn.1673-9736.2022.01.06>

LITERATURE SURVEY

❑ **Sierra Luoma et al. — Construction of a Fluxgate Magnetic Gradiometer for Integration with an Unmanned Aircraft System (Remote Sensing 2020)**

- Developed low-cost magnetic gradiometer, GPS receivers, Arduino-based data logger
- gradient mapping via field tests, detecting geological structures, metal objects
- prototype/proof-of-concept stage, not yet tested on known magnetic targets.

<https://www.mdpi.com/2072-4292/12/16/2551>

❑ **Li Y. et al. — A Magnetic Interference Compensation Method for Airborne Electronic Equipment without Current Sensors (Remote Sensing 2022)**

- Proposed a multi-source two-channel linear time-invariant (MTLI) model
- compensates for both maneuvering and OBE magnetic interference
- any in-flight mechanical shift invalidates the assumption

PROBLEM STATEMENT

CONTEXT:

- Magnetic interference **overwhelms signals needing's**

PROBLEM:

- Onboard electronics cause magnetic interference.
- Distorted sensor readings.

IMPACT:

- Navigation errors
- **Reduced UAV reliability** for subtle geological signals.



CAUSES

- **Permanent Magnetic Field** : Caused by the permanent magnetic moments of the **SYSTEM** (hard iron components) produce offset(unwanted magnetic value).
- **Induced Magnetic Field** : Caused by the **Earth's magnetic field changes** influenced by orientation
- **Current Field : Moving electric charges** .creates magnetic field

EXISTING SYSTEM

❖ GROUP1- Magnetometer Self-Calibration (Sensor Errors)

■ Hard & Soft Iron Calibration:

- **Hard Iron** : shift (Adds a **constant offset** to your readings)
- ✓ **Solution** : Offset correction) & Subtract it
- **Limitations:**
 - Cannot fix distortion (soft iron)
 - Cannot handle time-varying interference
- **Soft iron** : shape distortion (Distortion effect (not permanent magnet))
- ✓ **Solution**: Scaling + rotation correction
- **Limitations** :
 - Fails when motors are running (dynamic EMI)

https://web.stanford.edu/group/scpnt/gpslab/pubs/papers/Gebre_ICINS_2001_ins201.pdf

<https://ez.analog.com/mems/w/documents/4493/hard-soft-iron-correction-for-magnetometer-measurements>

<https://journals.sagepub.com/doi/full/10.1177/0020294019890627>

EXISTING SYSTEM

❖ GROUP 2 — Mathematical Fitting & Optimization (Static Calibration)

➤ Ellipsoid Fitting (Core Method):

- 3D magnetometer data collected during free rotation should produce a **perfect sphere**
- Creates Shifted data (hard iron) & Stretched / tilted data (soft iron)

✓ **solution** : Mathematical Model (Ellipsoid equation)

• Limitations :

- inefficient for real-time applications
- Changes shape as motor throttle changes, making a single pre-flight calibration invalid during dynamic operation.

<https://www.sciencepublishinggroup.com/article/10.11648/j.ijssn.20231101.13>

EXISTING SYSTEM

❖ GROUP 3 — Dynamic / Intelligent Methods

➤ Sensor Fusion (Kalman Filter / Complementary Filter)

- Fuses gyroscope, accelerometer, and magnetometer data using recursive state estimation to track orientation and filter out sensor noise.
- **Limitation:**
 - UAV to have a stable flight state
 - difficult to achieve, resulting in poor long-term reliability
 - it cannot model or subtract a structured, throttle-correlated magnetic noise source.

<https://journals.sagepub.com/doi/full/10.1177/00202940241227069>

<https://www.sciencedirect.com/science/article/abs/pii/S0168169916312418>

EXISTING SYSTEM

GROUP4- Passive techniques

❑ Hardware design — no external power needed

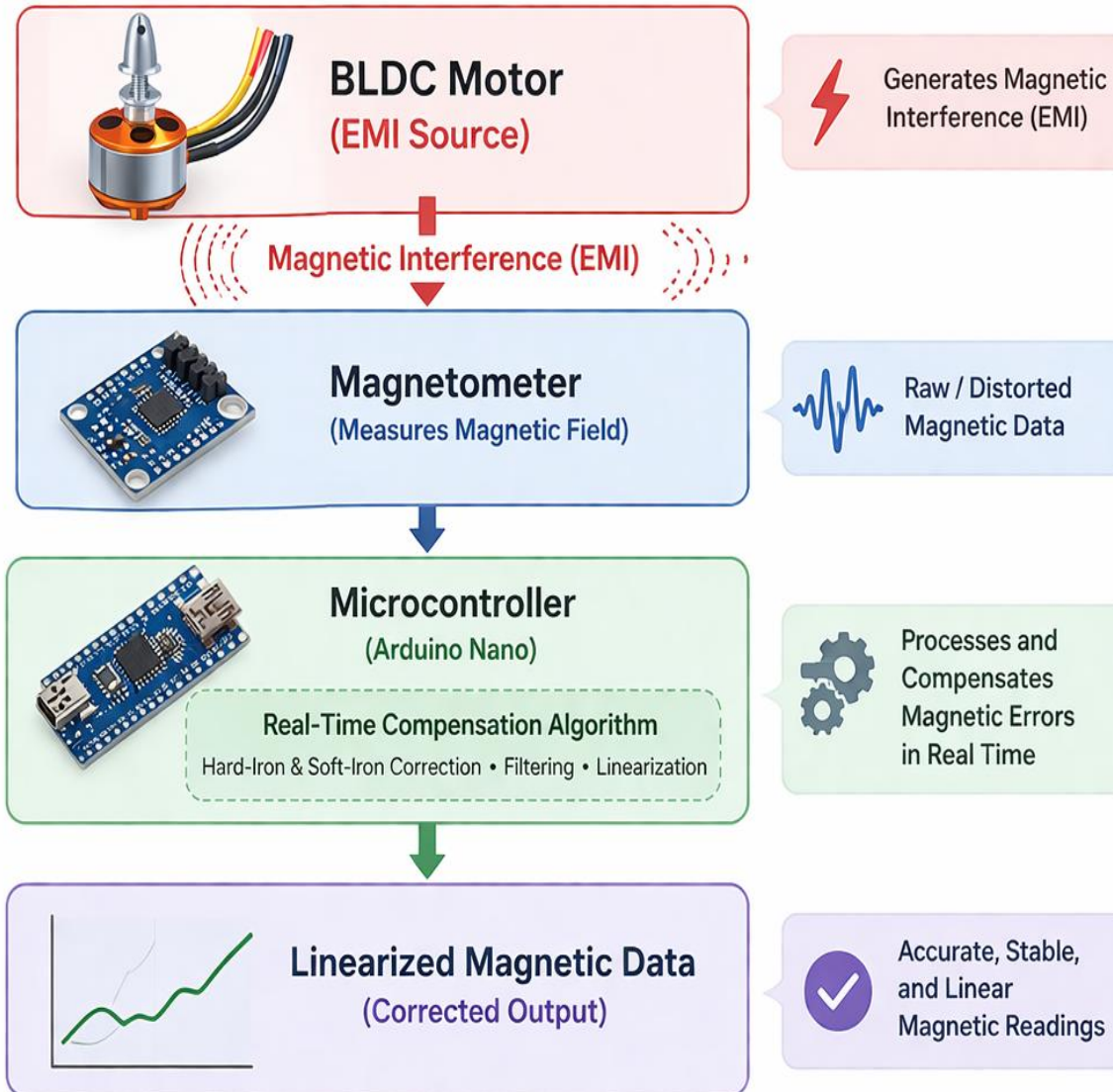
- Wiring separation — power cables away from signal lines
- Twisted wires — minimizes loop area & radiation
- Short signal paths — reduces noise pickup
- Star grounding — avoids ground loops
- Physical separation — motors/pumps away from sensors
- Ferrite beads & filters — suppresses high-frequency noise

GROUP4- Active techniques

❑ Dynamic methods when passive is insufficient

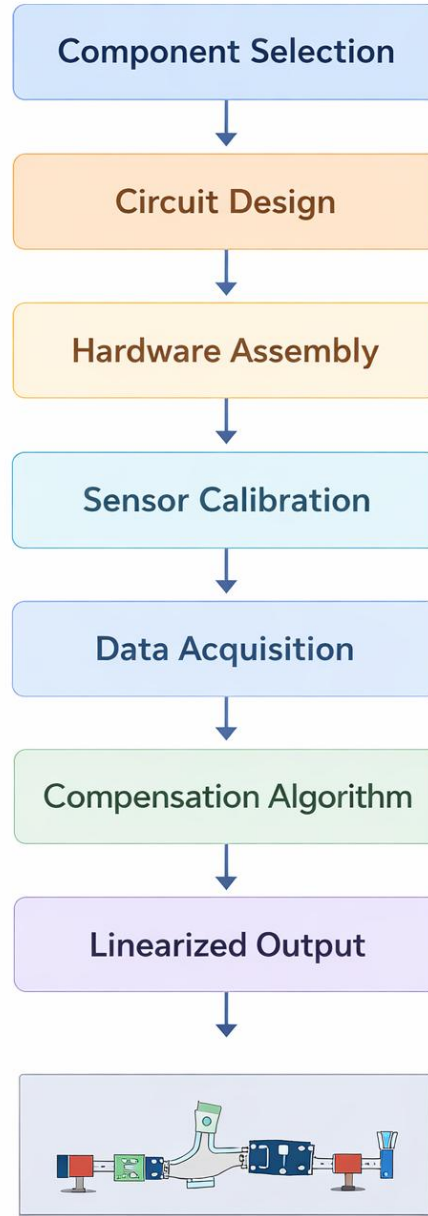
- Ground isolation — separates noisy power from sensitive electronics
- Optocouplers — signal transfer without electrical connection
- Signal isolation circuits — prevents noise propagation

Proposed System Overview



*PROPOSED
SYSTEM*

Methodology



METHODOLOGY

- **Components** selected based on **application**
- **Proper wiring** and interfacing ensured
- **Sensor rotated** in multiple directions
- **Magnetometer readings** (X, Y, Z) , Motor **ON** condition, **Interference** captured
- **Offset** removal
- **Linearization**
- Distorted → **corrected data**

COMPONENTS



Arduino Nano (ATmega328P)

- microcontroller board

Magnetometer (QMC5883L)

- 3-axis digital magnetometer .
- Measure the strength and direction of magnetic fields.



Current Sensor (ACS712)

measure current in the system{measure real-time motor current}



Li-Po Battery (3S, 11.1V, 2200mAh)

- provides the main DC power supply to the complete system

ESC (Electronic Speed Controller)

- receives power from the battery and regulates motor speed using PWM control signals



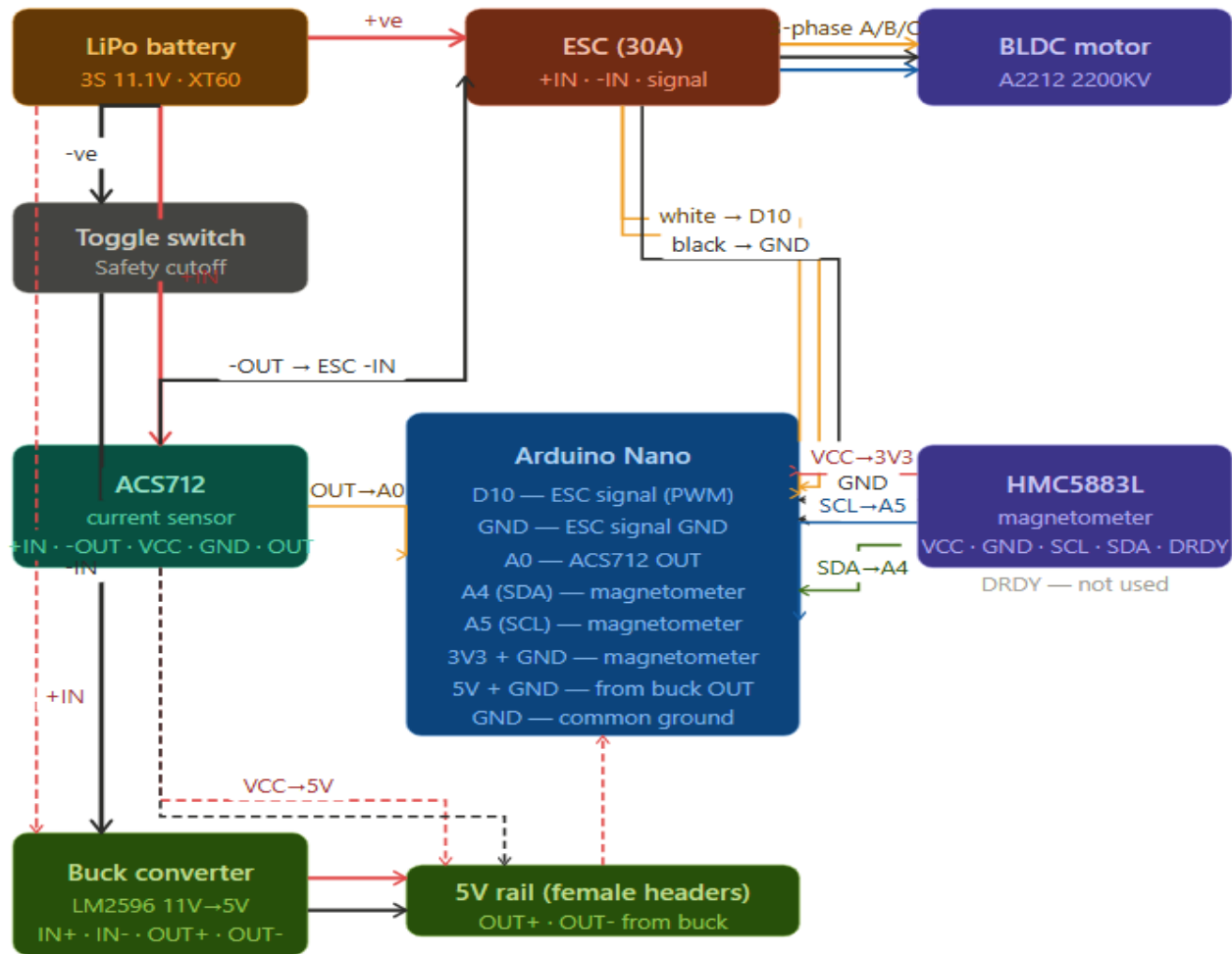
BLDC Motor (A2212)

- a brushless DC motor commonly used in drones (1000KV)

Buck Converter

- DC-DC step-down converter
- reduce voltage from a higher level to a lower level efficiently
- step down the battery voltage (11.1V) to a suitable level (5V)

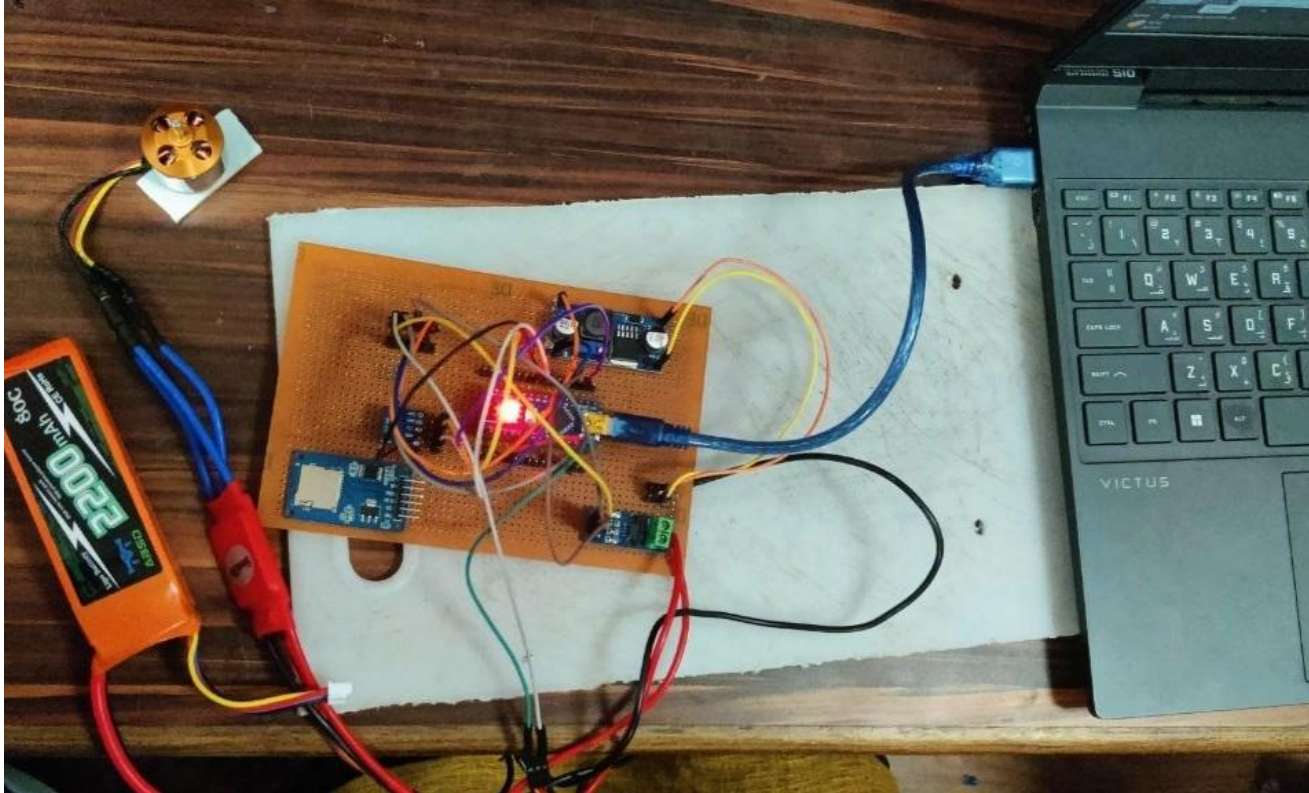




CIRCUIT DESIGN

Wire colour legend

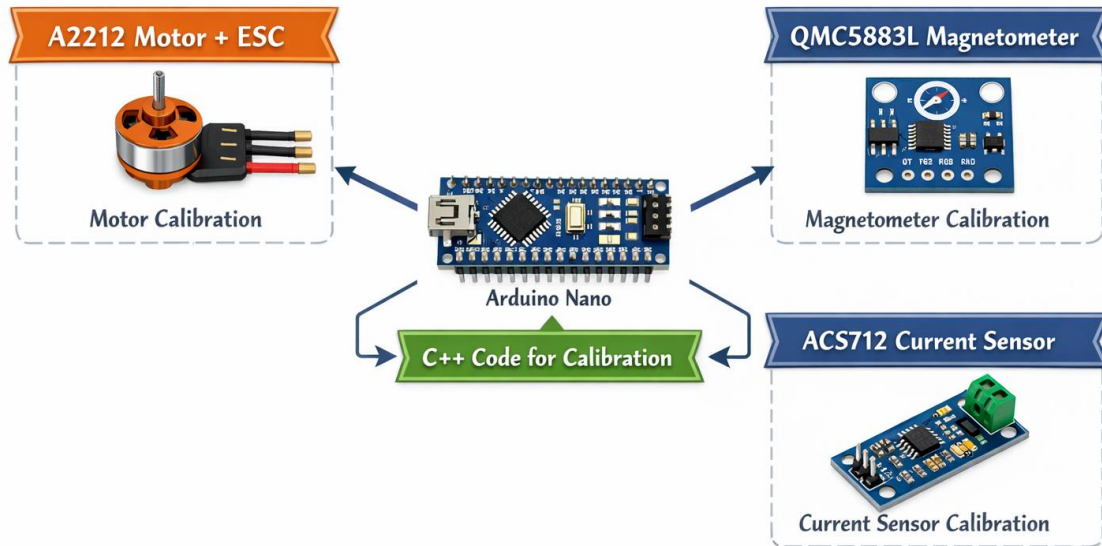
- Red (thick) — high-current +ve power (battery → sensor +ve IN) / ESC white signal → D10 (PWM) / ACS712 OUT → A0
- Black (thick) — high-current -ve GND (battery → toggle switch -ve IN) / BLDC - SCL → A5 (I2C clock) / ESC phase wire
- Red (dashed thin) — 5V power feed to sensors and Arduino
- Green — SDA → A4 (I2C data)



SYSTEM ASSEMBLY

*THE SYSTEM MEASURES MAGNETIC
DISTURBANCE FROM THE MOTOR,
PROCESSES IT USING A
MICROCONTROLLER, AND OUTPUTS
COMPENSATED LINEAR DATA.*

Components Calibration Flowchart



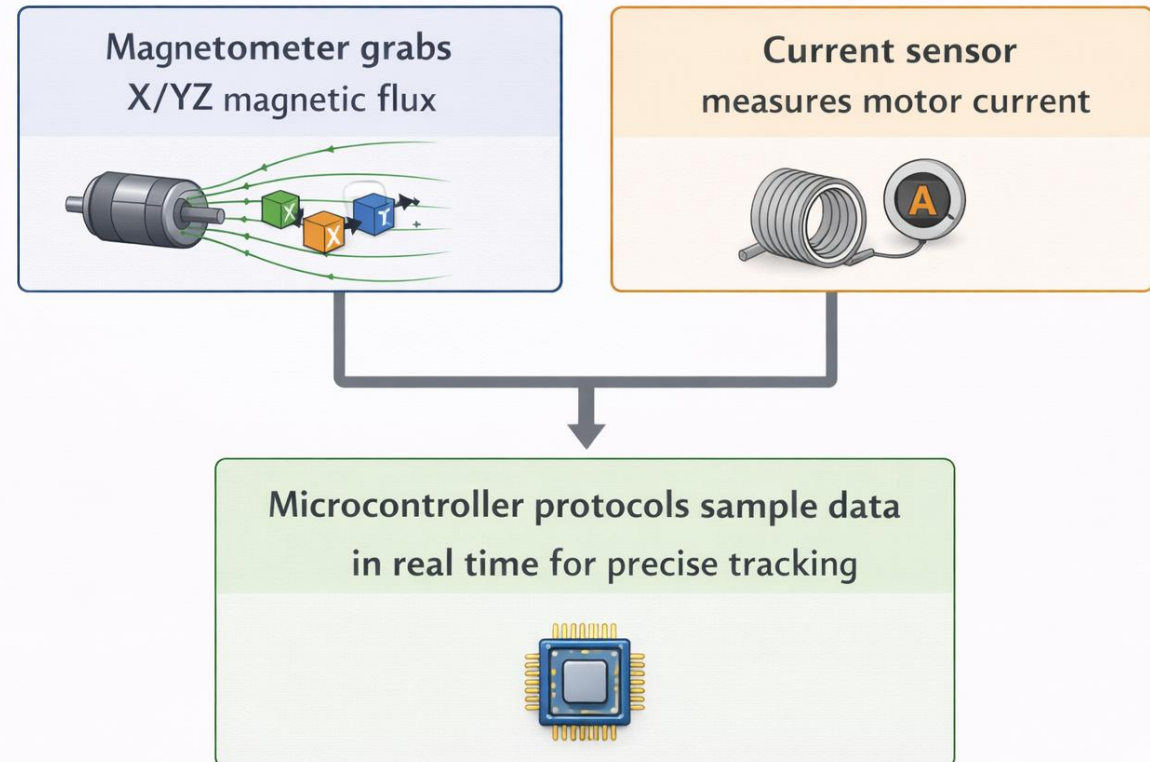
```
Log: Calibrating_Current...
Log: QMC5883L_Ready
Time (ms) , Speed, Amps, RawX, RawY, RawZ, CleanX, CleanY, CleanZ
4026,1000,-0.00,-23,771,456,-23.00,771.00,456.00
4269,1000,-0.00,-21,783,467,-21.00,783.00,467.00
4472,1000,-0.00,-23,768,467,-23.00,768.00,467.00
4676,1000,-0.00,-13,766,456,-13.00,766.00,456.00
4880,1000,0.15,-16,783,450,-16.15,782.85,449.85
```

CALIBRATION SETUP

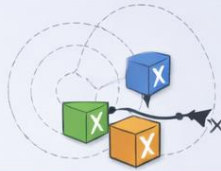
- Arduino IDE → C++ code → Arduino Nano
- Single firmware **calibrates all 3 components**
- **ESC arming + PWM profiling (1000–2000 μ s)**
- **0–12A no-load verification(current sensor)**
- **Idle Baseline: X/Y/Z axes sync (magnetometer)**
- **All sensors timestamp-aligned**

DATA PROCESSING

Acquisition – Raw Data Capture

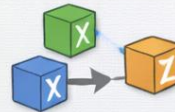


Compute mean offsets from
idle-state samples (*offset_{X/Y/Z}*)



Subtract: $r_x = x - \text{offset}_x$
(same for Y/Z)

Cleans ambient bias and removes sensor
drift upfront.



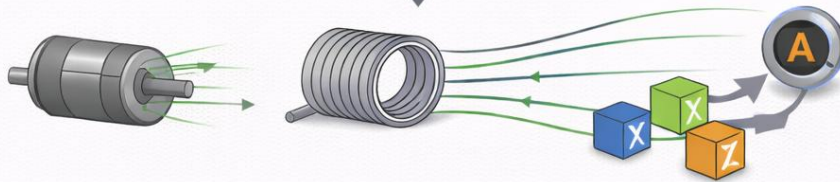
Offset Removal

Eliminate current-proportional noise:

$$C_X = r_X - (K_X \cdot I)$$

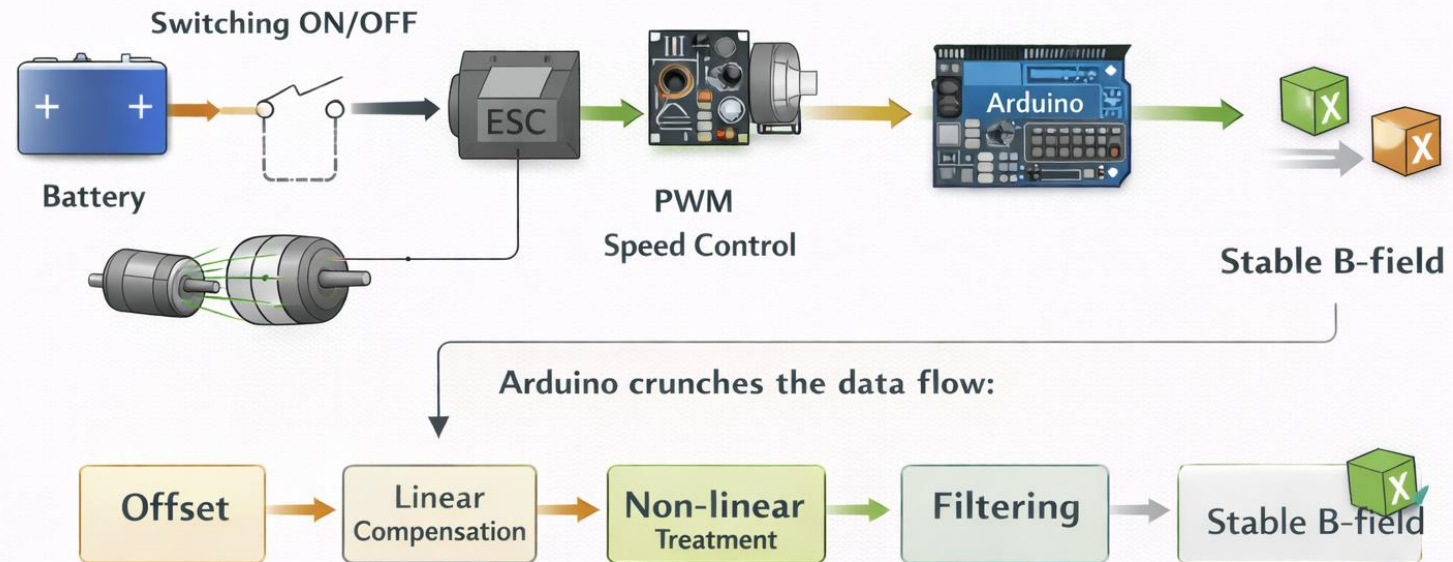
$(K_Y/K_X \text{ for Y/Z}).$

Directly cancels EM interference from motor and ESC current.



Linear Compensation

System Workflow – From Power to Clean Output



Battery powers **ESC** and motor; buck converter delivers stable 5 V to Arduino and sensors. Switching unit turns ON/OFF; ESC uses PWM for speed control; motor spinning creates EM noise.

Offset Correction

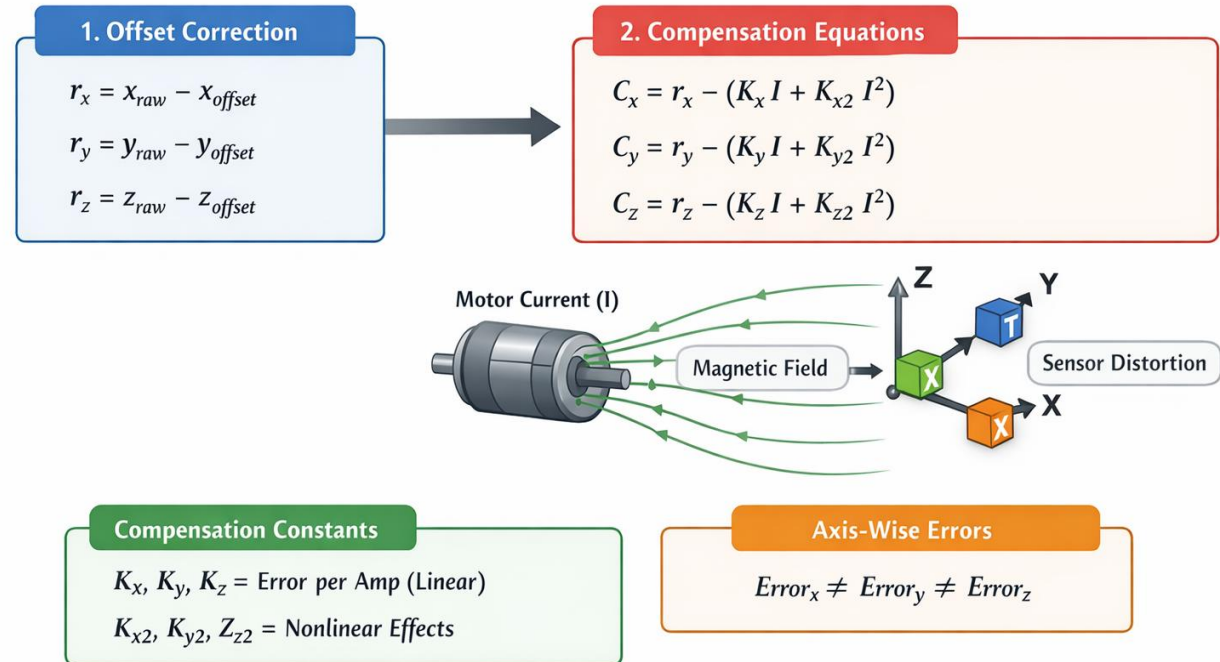
$$r_x = x_{\text{raw}} - x_{\text{offset}}$$

$$r_y = y_{\text{raw}} - y_{\text{offset}}$$

$$r_z = z_{\text{raw}} - z_{\text{offset}}$$

MATHEMATICAL MODEL

Magnetometer Current Compensation Model



❑ Magnetometer Current Compensation Model

Step 1: Offset Correction:

Were,

➤ **x, y, z** are **raw data**

➤ **X offset** similar to **y, z** are **baseline constant** (when motor is off)

Current → Magnetic Field → Axis Compensation

MATHEMATICAL MODEL

Step 2: Compensation Model

Where,

- K_x, K_y, K_z : Error per Amp (linear)
- K_{x2}, K_{y2}, K_{z2} : Nonlinear effects
- Current induces magnetic field → sensor distortion
- Disturbance varies by axis:

$$C_x = r_x - (K_x I + K_{x2} I^2)$$

$$C_y = r_y - (K_y I + K_{y2} I^2)$$

$$C_z = r_z - (K_z I + K_{z2} I^2)$$

$$\text{Error}_x \neq \text{Error}_y \neq \text{Error}_z$$

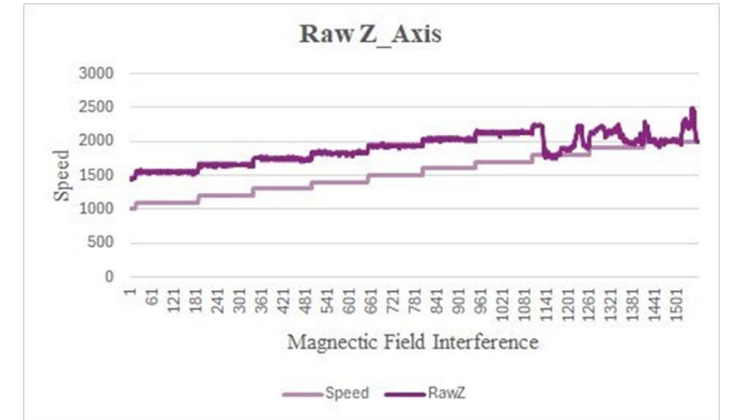
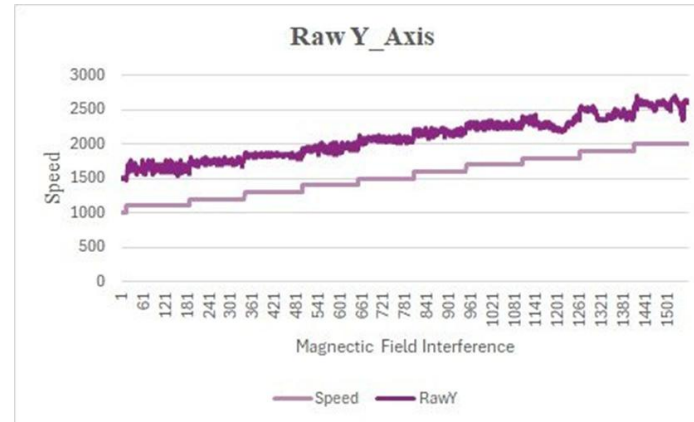
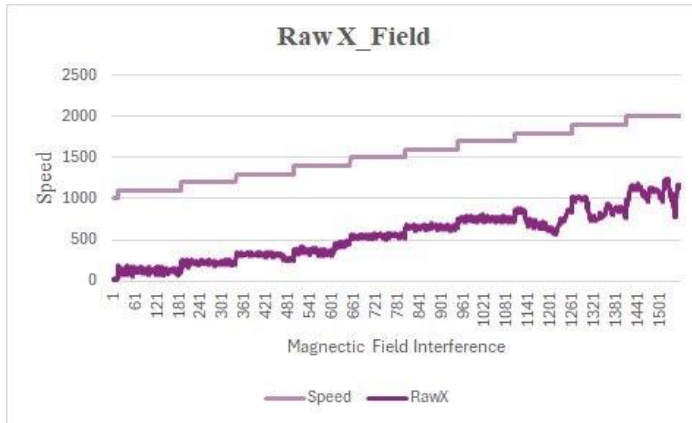
➤ **Motor Current → Magnetic Field → Axis-wise Error → Compensation**

MAGNETIC INTERFERENCE FROM MULTICOPTER

II



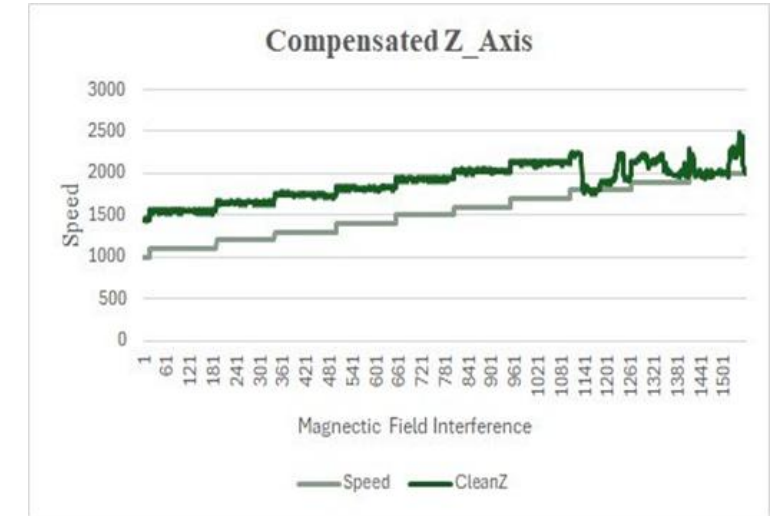
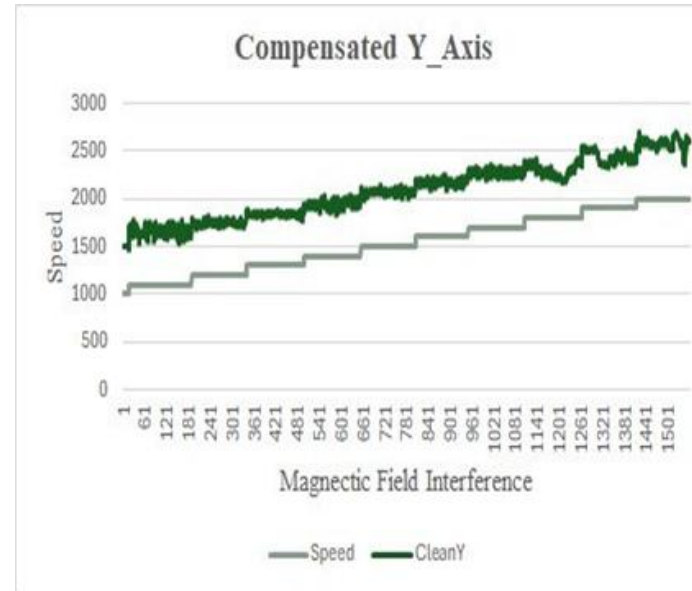
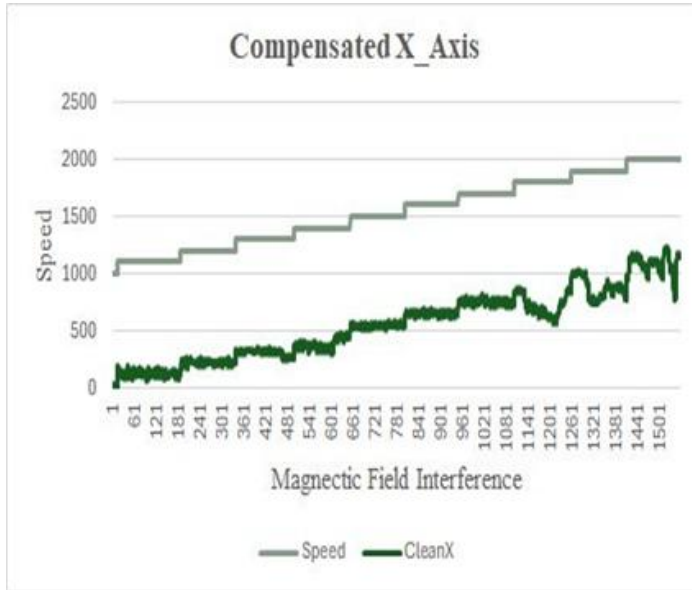
Trial-1 Raw Magnetic Field (X, Y, Z)



RESULT

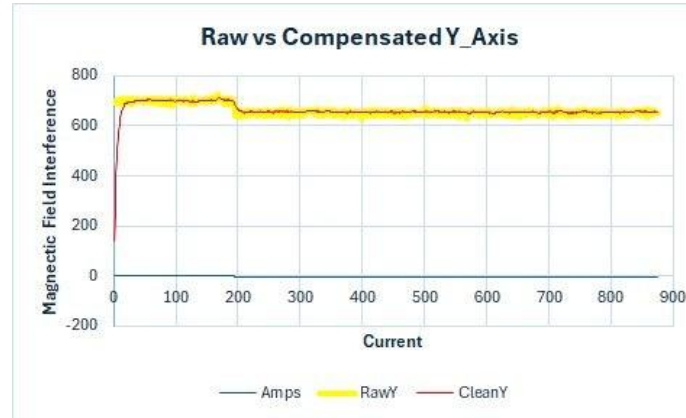
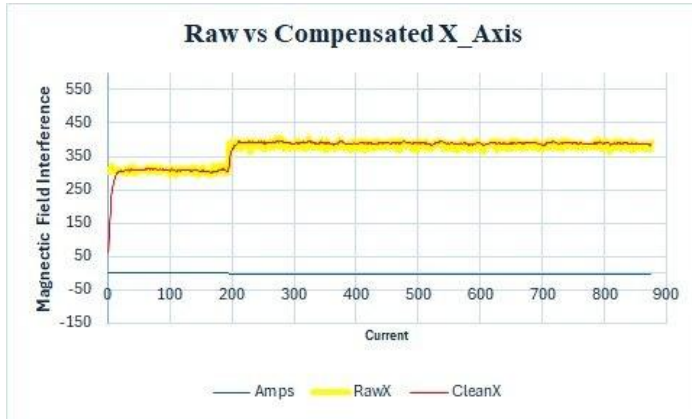
RAW VS COMPENSATION OF X-AXIS

Trial- 2 Compensated Magnetic Field (X, Y, Z)



RESULT

Trial- 3 Compensated Magnetic Field (X, Y, Z)



RESULT

```
Serial.println(cz);
} else {
  Serial.print("RawX:"); Serial.print(rx); Serial.print(" ");
  Serial.print("CleanX:"); Serial.print(cx); Serial.print(" ");
  Serial.print("RawY:"); Serial.print(ry); Serial.print(" ");
  Serial.print("CleanY:"); Serial.print(cy); Serial.print(" ");
  Serial.print("RawZ:"); Serial.print(rz); Serial.print(" ");
  Serial.print("CleanZ:"); Serial.println(cz);
}

// Speed control
if (Serial.available()) {
  int input = Serial.parseInt();
  if (input >= 1000 && input <= 2000) {
    speed = input;
    esc.writeMicroseconds(speed);
  }
}

delay(200);
}
```

Serial Monitor X

pt (Enter to send message to 'Arduino Nano' on 'COM4')

850.00	CleanX:-867.66	RawY:760.00	CleanY:739.23	RawZ:408.00	CleanZ:419.22
863.00	CleanX:-867.66	RawY:760.00	CleanY:742.75	RawZ:413.00	CleanZ:417.31
876.00	CleanX:-870.29	RawY:762.00	CleanY:745.94	RawZ:386.00	CleanZ:410.85
888.00	CleanX:-874.84	RawY:731.00	CleanY:742.27	RawZ:450.00	CleanZ:417.55
892.00	CleanX:-871.30	RawY:766.00	CleanY:746.32	RawZ:370.00	CleanZ:407.30
898.00	CleanX:-875.54	RawY:740.00	CleanY:744.44	RawZ:391.00	CleanZ:403.39
940.00	CleanX:-869.37	RawY:713.00	CleanY:737.51	RawZ:500.00	CleanZ:422.04
971.00	CleanX:-870.69	RawY:745.00	CleanY:738.34	RawZ:400.00	CleanZ:418.52
966.00	CleanX:-870.73	RawY:747.00	CleanY:739.41	RawZ:400.00	CleanZ:414.11
966.00	CleanX:-870.85	RawY:740.00	CleanY:738.80	RawZ:436.00	CleanZ:417.72
973.00	CleanX:-872.40	RawY:747.00	CleanY:739.68	RawZ:456.00	CleanZ:424.57

RESULT

CONCLUSION

The system non-linearity is minimized by estimating the offset through a **mathematical model** **Motor Current → Magnetic Field → Axis-wise Error → Compensation**.

The achieved compensation levels are:

- **X-axis: 28%,**
- **Y-axis: 46%,**
- **Z-axis: 27%,**
with an **overall system compensation of 54%**, demonstrating effective error reduction and improved linearity.

LIMITATIONS

- Assumes linear relationship between current and magnetic interference
- Limited effectiveness for non-linear conditions
- Requires manual tuning of coefficients (K_x , K_y , K_z)
- Performance affected by sensor noise and spikes
- Tested under limited motor speed conditions
- Uses single magnetometer (no sensor fusion)

APPLICATION

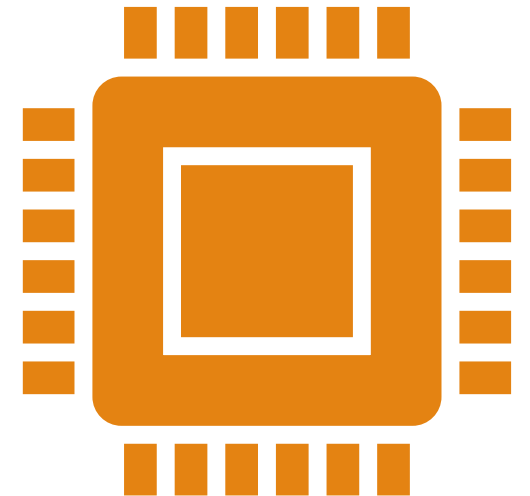
- **UAV Navigation:** Boosts magnetometer heading accuracy for stable flight and precise control.
- Airborne magnetic surveys
- **Drone Mapping:** Sharpens orientation for surveying, mapping, and environmental monitoring
- **Autonomous Drones:** Ensures reliable sensor data for flight control algorithms.
- **Robotics/Embedded:** Compensates interference in motor-proximal magnetometer setups.
- **R&D:** Enhances sensor performance in UAV/experimental systems.

FUTURE SCOPE

- Adaptive compensation for dynamic operating conditions
- Machine learning for better interference prediction
- GPS & IMU integration for advanced drone navigation
- Wireless real-time monitoring interfaces
- Testing across diverse environmental conditions
- Miniaturization for compact UAV/embedded systems

SUMMARY

- **Developed a prototype** system to analyze and compensate motor-induced magnetic interference in UAVs
- **Implemented using embedded systems** and sensor calibration techniques
- Achieved **linear system** in magnetic interference under test conditions
- System shows potential for further optimization and research development
- **Future Scope** Enhance algorithm accuracy, Improve real-time performance Potential for research publication and advanced UAV applications



REFERENCE

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